

Product Data Exploration using Microsoft Excel

Objective:

To analyse and explore product data using Microsoft Excel by applying different formulas and functions for calculations, categorization, and text extraction.

Tasks Performed:

- Calculated the total price, number of products, and average product price using Excel formulas.
- Identified the minimum and maximum product prices in the dataset.
- Created a new Price Range column using the IF function to classify products as:
 1. High Price (Price $\geq$ \$500)
 2. Standard Price (Price $<$ \$500)
- Used SUMIF to calculate the total price of products in the Electronics category.
- Used COUNTIF to count products priced below \$100.
- Applied text functions on Product ID to extract:
 1. Day using LEFT
 2. Month using MID
 3. Country Code using RIGHT
- Organized the worksheet with clearly labeled columns and calculated outputs.

Formulas Used:

- =SUM(D2:D35) – Total price calculation
- =COUNTA(B2:B35) – Total number of products
- =AVERAGE(D2:D35) – Average product price
- =IF(D2>500,"High Price","Standard Price") – Price range classification
- =LEFT(A2,2) – Extract Day from Product ID
- =RIGHT(A2,2) – Extract Country Code from Product ID
- =MID(A2,4,3) – Extract Month from Product ID

Findings:

1. Sum, Count and Average:
 - Total Price of all products = \$ 10,100
 - How many products exist = 34
 - Average price of the products = 297.05
2. Minimum and Maximum Values:
 - Minimum price of all products = 30
 - Maximum price of all products = 1000
3. Conditional Functions – SUMIF and COUNTIF:
 - Total price of products in Electronics category
SUMIF = 8050
 - Number of products with a price less than \$100.
COUNTIF = 11

Conclusion:

This project improved my basic Excel and data analysis skills by using formulas and functions to organize, analyse, and extract useful information from the dataset.